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The Undergraduate Research Journal of Sup'Biotech

Volume 6 | February 2023

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## Additional Information

This issue of *Sup'Biotech Projects* contains research articles written by students enrolled in the Apprenticeship engineering program at Sup'Biotech, as part of their contribution to the Undergraduate Research Program in Bioengineering (5D Projects), supervised by Julien Grimaud, PhD. For more information, please contact Dr. Grimaud using the following address: julien.grimaud [at] supbiotech.fr

# Editorial

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Welcome to Volume 6 of *Sup'Biotech Projects*! This issue is a collection of student research papers written by the graduating Apprenticeship class of 2023, as part of their contribution to the Undergraduate Research Program in Bioengineering, also known as 5D Projects.

The 5D Projects consist of transversal research and development projects in biotechnology, which accompany the Apprenticeship students throughout their stay at Sup'Biotech. This program revolves around the 5D methodology (Discover, Define, Design, Develop, Deploy). Each project is divided into four steps:

- 1) In groups, students write a scientific review on a topic of their choice related to biotechnologies. They then come up with ideas of innovative products and solutions.
- 2) Students assess the economic viability of their idea through extensive market research.
- 3) Then, as part of their proof-of-concept, students are given full access to Sup'Biotech Teaching Laboratories for two weeks to collect preliminary scientific data. Prior to entering the laboratory, each group must design their own scientific approach, write protocols, and manage their consumable stock using the budget allocated by the school.
- 4) Finally, each group writes a scientific article that summarizes all the information gathered throughout their 5D Project, with a focus on the preliminary data.

Each paper has been peer-reviewed by two separate groups of Sup'Biotech students, to ensure the highest level of quality for each submission. We are proud to present the fruit of their labors and hope that this journal will be a valuable resource for future students and researchers in the various fields of biotechnology. We are proud of their work and the

contributions they have made to the scientific community. We hope you enjoy the journal and will share its contents with others.

We would like to thank everyone involved in the creation of this journal, from the students and professors to the administration, for their hard work and dedication, including:

- The Office of Apprenticeship Students Affairs (Mathilde Savoie, Léa Ramfos) for their unconditional support;
- The Teaching Laboratories team (Estelle Mogensen, PhD, Baptiste Favard, Corentin Rousseau, Sofiane El Ouariachi, Roxane Loyant, Jacqueline Bert) without whom the proof-of-concept would have not been possible;
- Claire Grassi, for her invaluable help and support during the market research phase of the 5D Projects;
- All members of the Research Laboratories at Sup'Biotech, who proposed the initial research topics and helped our students at various stages of their project.
- The students who submitted cover designs. Congrats to the winning submission (prodigiosin group), which earns to be the “spotlight” article of this issue!

Last but not least, our warmest thanks go to the Apprenticeship class of 2023. As the inaugural “5D Projects” class, you played a pivotal role in designing this new undergraduate research program, through your willingness to do hard work and give constructive feedback. You have worked hard and accomplished so much. You should feel proud of your achievements and the amazing milestone you have reached. Wishing you all the best in your future endeavors. Cheers to your success!

# A Platform for the Transformation of Bacteria to Produce Nanobodies

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**Abstract** – Alzheimer disease is a neurodegenerative disease that affects memory, thinking, and behavior. It affected 1.2 million people in France and will represent 2 million patients in 2030. There is no cure for Alzheimer's disease, but treatments can help to manage symptoms and improve quality of life if taken care of at an early stage. However, definitive diagnosis can only be made by a combination of tests leading to a late diagnosis. The glial fibrillary acidic protein appears to be a good biomarker of neurological lesions and neurodegenerative diseases as it is observed following a stroke activating the gliosis mechanism in the hours after the lesion. Nanobodies that can spontaneously cross the blood brain barrier has led to a growing field of research into their potential uses and applications including in detection. Consequently, producing nanobodies targeting GFAP could represent an answer to many patients worldwide and thus improve drastically their medical care thanks to an early diagnosis. This article will present how to design a platform for large-scale nanobodies production. We obtained colonies of DH5- $\alpha$  that may contain nanobodies targeting GFAP. Further studies should be done to verify that we obtained the nanobodies to confirm the results.

**Index Terms** – Bacteria transformation, biomarker, GFAP, nanobody, neurodegenerative disease, plasmid.

## INTRODUCTION

A neurodegenerative disease is a type of illness that causes a gradual loss of function of nerve cells in the brain and spinal cord [1]. These diseases can lead to problems with movement, memory, and thinking. Some examples of neurodegenerative diseases include Alzheimer's disease, Parkinson's disease, and Huntington's disease [1]. They are caused by a variety of factors, including genetic mutations, environmental toxins, and lifestyle factors [1].

Alzheimer's disease is a progressive neurodegenerative disease that affects memory, thinking, and behavior [2]. It is the most common cause of dementia, a general term for a decline in cognitive function severe enough to interfere with daily life. Symptoms of Alzheimer's disease typically develop gradually and worsen over time, eventually leading to a loss of the ability to carry out the simplest tasks. There is no cure for Alzheimer's disease, but treatments can help to manage symptoms and improve quality of life [2]. Alzheimer disease represents up to 60 to 80% of all dementia cases in the world, around 24 million people are affected worldwide by this disease [2]. In France, it is estimated that around 1.2 million people have Alzheimer's disease and other forms of dementia. This number is projected to increase to 2 million by 2030 [3].

There are several methods that can be used to diagnose Alzheimer's disease, including:

- Cognitive and neuropsychological assessments: These tests assess memory, language, attention, and other cognitive abilities. They can help to identify specific areas of impairment and can provide a baseline for monitoring changes over time [4].
- Brain imaging: Techniques such as magnetic resonance imaging (MRI) and positron emission tomography can help to identify changes in the brain that are characteristic of Alzheimer's disease, such as shrinkage of the hippocampus [4].
- Lumbar puncture: A lumbar puncture is a procedure in which a small amount of cerebrospinal fluid (CSF) is removed from the spinal column using a needle. This fluid surrounds the brain and the spinal cord and can be analyzed for biomarkers that are associated with Alzheimer's disease. Even though lumbar puncture is a relatively safe procedure, it still is an invasive method causing many downsides such as pain [4].

It is important to note that while these tests can be helpful in identifying Alzheimer's disease, they are not conclusive, and a definitive diagnosis can only be made by a combination of cognitive tests, medical history, physical examination, and other diagnostic tools. These tests are not done at an early stage because symptoms have already appeared and thus lead to a late diagnosis [4].

The glial fibrillary acidic protein (GFAP) is a structural protein expressed in many cells in the brain named glial cells and produced in larger quantities in Alzheimer disease. The glial cells have a role in the protection, transmission, nutrition, and support of neurons, and unlike neurons they can divide by mitosis [5]. During brain trauma they proceed to fill the loss of nervous tissue. This healing phenomenon is called gliosis. Indeed, an overexpression of GFAP is observed following a stroke activating the gliosis mechanism in the hours after the lesion [6].

In addition, neurodegenerative diseases can also affect the expression, phosphorylation or splicing of the protein [7][8]. GFAP therefore appears as a good biomarker of neurological lesions and neurodegenerative diseases. The visualization of the protein by immunostaining could allow the detection of the diseases at early stages. However, the blood-brain barrier is one of the most selective barriers to cross in the human body [9].

In neurodegenerative diseases, like Alzheimer's, there is a progressive loss of neurons, which leads to a decrease in the number of neurons in the brain. This can trigger a cascade of events that leads to the activation of astrocytes, which respond by increasing their GFAP content. Elevated levels of GFAP in the cerebrospinal fluid (CSF) and/or brain tissue have been found in people with Alzheimer's disease and other forms of dementia and are considered as a biomarker for these diseases [4].

Monoclonal antibodies are mainly produced by rodent cells which can cause an immune reaction when they are injected into a human patient. To counter this problem, monoclonal antibodies must be chimeric or humanized, which requires other treatments and more expenses [10]. Compared to conventional monoclonal antibodies, nanobodies have a single domain but with the same antigen-binding specificity. They are able to bind specifically to antigens, even in places that are very difficult to access like the blood-brain barrier thanks to their small size and high solubility [11].

Nanobodies are small, stable, single-domain antibodies (fig.1) that are derived from heavy-chain only antibodies found in camelids (such as camels and llamas). They are being explored for use in a variety of applications, including diagnostics, therapeutics, and imaging. Thanks to their small sizes, nanobodies can spontaneously cross the blood brain barrier. This particularity has led to a growing field of research into their potential uses and applications [12].

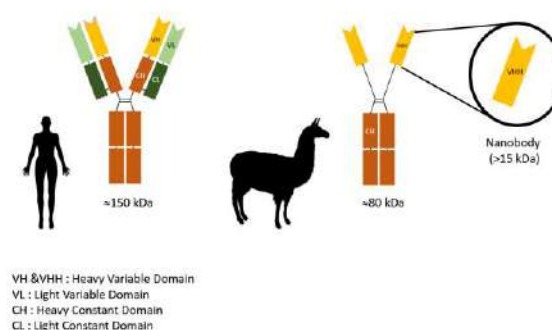


FIGURE 1 – COMPARAISON OF A NANOBODY AND AN ANTIBODY

This figure shows the difference between a classic antibody and a nanobody in terms of structure and size. A classic antibody consists of 2 heavy chains and 2 light chains and weighs approximately 150 kDa. In comparison, the camelid antibody weighs approximately 80 kDa and is composed of only 2 heavy chains with a heavy variable domain (VHH) that constitutes the nanobody. The nanobody is 10 times smaller than the conventional antibody which makes it very interesting for many uses in medicine.

As seen previously, the detection of Alzheimer's disease is based on a full battery of tests, which are a very long and painful. Producing nanobodies targeting GFAP could represent an answer to many patients worldwide and thus improve drastically their medical care thanks to an early diagnosis.

Because platforms for large-scale nanobodies production do not exist yet, we tried to answer here if it is possible to create a plasmid multiplication platform in order to produce nanobodies targeted against GFAP ? Firstly, our goal is to design a protocol for the transformation of bacteria to produce nanobodies against GFAP. To do so, designing a protocol is essential to reach this aim.

## MATERIAL AND METHOD

### I. SUBCULTURES AND CULTURES OF BACTERIA

#### Subculture and Culture of *E.Coli* DH5- $\alpha$

DH5 $\alpha$  is an *E. coli* strain that is commonly used for plasmid amplification. First, we tried to construct our recombinant plasmid by inserting the coding sequence of nanobodies against GFAP into pET-22 plasmid. We digested the plasmid pET-22 with two restriction enzymes: Not1 (New England Biolabs, R0189S) and Xho1 (New England Biolabs, R0186S). After an electrophoresis for the digestion control, we didn't find the insert on the gel. We decided to continue our experience by using DH5- $\alpha$  bacteria from A. Leleu which contained the recombinant plasmid (pET-23+ coding sequence of nanobodies targeted GFAP).

DH5- $\alpha$  bacteria were used to contain the recombinant plasmid pET23 (which contains the coding sequence for nanobodies targeted against GFAP). *E.Coli* DH5- $\alpha$  were cultivated with LB (Lysogeny Broth) media : 1g of yeast extract (Sigma-Aldrich, 70161), 2g of Trypton (Sigma-Aldrich, T7293), 2g



of NaCl (SIGMA, S8776) and ampicillin 50mg/mL ((Sigma-Aldrich, A1000000) in 200mL distilled water). For the subculture, an aliquot of bacteria in 100ml of LB media with ampicillin were cultivated at 37°C 150 rpm during 16h. To obtain a sufficient concentration of the bacteria of interest, 1,5mL of the subculture were inoculated in LB media with ampicillin until we get a DO of 0,4-0,5. In order to isolate the plasmid, the amplification bacteria were plated on a petri dish with LB medium supplemented with ampicillin to isolate a colony.

### Subculture et culture BL21

Cell BL21 is an *E. coli* bacterial strain chemically competent to express high level proteins using an RNA polymerase-IPTG T7 induction system.

*E. coli* BL21 (provided by PI Lafaye) were cultivated with LB media : 1g of yeast extract (Sigma-Aldrich, 70161), 2g of Trypton (Sigma-Aldrich, T7293), 2g of NaCl (SIGMA, S8776) in 200mL distilled water. For the subculture, an aliquot of bacteria in 100ml of LB media were cultivated at 37°C 150 rpm during 16h. To obtain a sufficient concentration of the bacteria of interest, 1,5mL of the subculture were inoculated in LB media until we get a DO of 0,4-0,5.

### Competent BL21

In order to make the BL21 competent CaCl<sub>2</sub> 0,1M (Sigma-Aldrich, C3306-100G) bacteria was added and kept in ice for 10 minutes before centrifuging at 4000 rpm for 5 minutes at 4°C. The pellet was resuspended with 10mL of cold CaCl<sub>2</sub> 0,1M (Sigma-Aldrich, C3306-100G) and vortexed vigorously.

In order to isolate the plasmid, the amplification bacteria were plated on a petri dish with LB medium supplemented with Ampicillin (Sigma-Aldrich, A1000000), to isolate a colony in order to cryopreserve it.

## II. USE OF BACTERIA OF INTEREST

### Plasmid purification

The ZymoPURE™ Plasmid Miniprep Kit (ZymoResearch, D4208T) was used for the purification of the plasmid after the culture of selected bacteria before putting it into the expression bacteria. We followed the supplier's protocol, ZymoResearch to perform this purification [13].

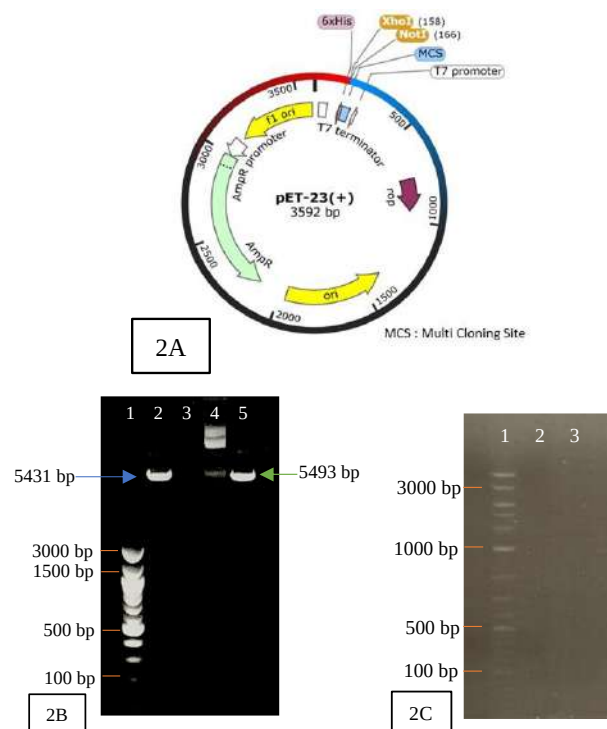
### Gel Migration

To validate the plasmidic construction a gel electrophoresis was performed. The gel used was a 1% agarose gel (Sigma Aldrich, 1.16802) in TAE buffer (TrisBase, acetic acid, EDTA, H<sub>2</sub>O) in addition to the Ethidium Bromide (Sigma Aldrich, E1385). The two samples were prepared by adding 25µL of double digested solution or water for the control sample, 3µL of DNA Gel Loading Dye (6X) (ThermoFisher Scientific, Laemmli, 15492859) and 3µL of water for 30µL sample. After that, 20µL of the samples and the molecular

weight marker were loaded, and migration was conducted between 150V depending during 1h30.

### Cryopreservation

For the cryopreserving cells, sterile glycerol 50% (Sigma Aldrich, G2025) was used and kept in cryotubes at -80°C.



**FIGURE 2 – ELECTROPHORESIS 1% AGAROSE GEL WITH THE RESTRICTION MAP (MADE WITH SNAPGENE® SOFTWARE) CORRESPONDING TO THE PLASMID OF INTEREST**

(A) **Restriction map.** The figure 2C represents the restriction map of pET-23, our plasmid of interest. The 2 enzymatic restriction sites are: XhoI and NotI.

(B) **Results of an electrophoresis gel.** The figure 2A is a photograph of the first 1% agarose electrophoresis gel, read under UV, aimed at verifying that the plasmid is present in the tubes. 1: Molecular weight marker (100 bp), 2: pET-22 after double digestion (5431bp), 3: Insert after double digestion (364 bp), 4: Native plasmid (coiled, super-coiled and relaxed), 5: Monodigested Plasmid (5493 bp).

(C) **Results of an electrophoresis gel.** The figure 2B is a photograph of the second 1% agarose electrophoresis gel, read under UV, in order to verify the presence of insert in our tubes and therefore validating our manipulation. 1: Molecular weight marker, 2: Contents of double digestion of the pET-23, 3: Negative control with water instead of plasmid.

## RESULTS

Following the above protocols, the following results were obtained. In the figure 2A, the bands of the column 1 correspond to the ladder of the molecular weight marker (100 bp). Identified in blue, in column 2, we observe the presence of a band at approximately 5431bp, which corresponds to pET22 after the double digestion. In column 3, no band is observable, the expected insert being 364 bp. The fragment of interest could therefore not be recovered. In

column 4, 3 distinct bands are observable; they correspond to the 3 native forms of the plasmid: coiled, supercoiled and relaxed. Identified in green in the fifth column, a band is observed at 5493 bp which corresponds to the monodigested plasmid. In the figure 2B, the bands in column 1 correspond to the molecular weight marker ladder (2log DNA ladder). In column 2 and 3 no band is observable. Probably due to faulty hardware. In the figure 2C is noticeable that the pET-23 contains a resistance gene for the ampicillin and one restriction site for NotI and XhoI.

Ampicillin an antibiotic allowing the selection of DH5- $\alpha$  with resistance against ampicillin. The figure 3 shows four colonies of DH5- $\alpha$  on the agar. In order to perform the pET-22 amplification, we transformed the bacteria and plated them on LB Lysogeny agar plates containing ampicillin. This allowed us to select DH5- $\alpha$  bacteria containing the recombinant plasmid with the nanobody sequence which also contains the ampicillin resistance. Controls were performed with bacteria that do not contain the ampicillin resistance gene to ensure that the selected bacteria contained our nanobodies sequence. Fig. 3 shows the presence of DH5- $\alpha$  colonies that allowed us to amplify our plasmid. We can see four colonies. DH5- $\alpha$  bacteria does not allow the production of nanobodies. We therefore used another strain of bacteria BL21 to product them.



**FIGURE 3 – SELECTION OF DH5- $\alpha$  CONTAINING PLASMID CULTURE OF DH5- $\alpha$  ON LB NUTRIENT AGAR CONTAINING AMPICILLIN**

Results of cultures of transformed DH5- $\alpha$  bacteria on Petri dish with ampicillin. The dish was poured with agar LB + Ampicillin and was then plated with transformed DH5- $\alpha$  bacteria. Circles show CFU (colony forming units).

The figure shows an absence of transformed BL21. The Fig. 4 shows the culture of BL21 transformed with the plasmid containing the nanobodies. This strain allows the production of nanobodies. However, we can see on the figure no bacteria grew on the agar. So, we could not produce nanobodies.

We carried out a culture of BL21 without the plasmid containing the ampicillin gene resistance in order to verify that only bacteria possessing the plasmid can grow in LB agar containing ampicillin. This agar is our negative control. We can see in figure 5A that no bacteria has grown, which shows

that only bacteria containing our plasmid coding for nanobodies with the resistance gene will be able to grow. We performed a second control with bacteria without plasmid on a non-selective agar to verify that these bacteria can grow. We can see in figure 5B that some bacteria were able to grow on this control agar.



**FIGURE 4 – RESULTS OF THE TRANSFORMATION OF THE BL21 CULTURE OF TRANSFORMED BL21 ON LB NUTRIENT AGAR**

Results of cultures of transformed BL21 bacteria on Petri dish. The dish was poured with agar LB and was then plated with transformed BL21 bacteria.

## DISCUSSION

There is a high interest in developing a method that uses the benefits of the nanobody and its properties for an early detection of neurodegenerative diseases. As a matter, such an early detection is not available on the market yet, regardless of a real need for it. Moreover, actual diagnosis is long and painful leading to a later treatment. Therefore, the aim of this study was to elaborate a protocol to transform bacteria to produce nanobodies and have a platform for larger scale production that can be later used to diagnose Alzheimer disease among other brain-related issues.

Concerning the results, we started by doing an electrophoresis to make sure that our plasmid and insert were present in the tubes. As we can observe in figure 2A, the plasmid was present in the first tube, but we did not recover the insert which was the key fragment for the next steps of our experiment. Due to this, we decided to change our plasmid to have one which already contains the insert. We verified it by the electrophoresis in the second electrophoresis (fig. 2B), but we were not able to observe any band under UV light which could be justified by faulty hardware. In fact, we noted afterwards that the kit of purification used (ZymoPURE™ Plasmid Miniprep Kit) was outdated, which has surely had negative consequences on our results. The culture of DH5- $\alpha$  allowed us to amplify the plasmid containing the nanobodies. However, the growth of the bacteria was weak as we could see on the petri dish. In fact, only four colonies managed to proliferate (fig. 3). This results in a low yield of plasmid, not very significant. This could be explained by different factors, such as the ampicillin that was used to select bacteria

containing the recombinant plasmid of our nanobodies: The antibiotic may have made it more difficult for the bacteria to multiply. Some bacteria could also have gotten rid of their plasmid and therefore of their resistance to ampicillin. We could see that the results of the transformation of BL21 were also not conclusive (fig. 4). The bacteria could not grow on the agar. This could be explained by the fact that the bacteria did not survive the thermal shock during the transformation. It is also possible that the bacteria at the beginning were not in good health. Finally, our negative results could also be justified by the use of the outdated kit of purification (ZymoPURE™ Plasmid Miniprep Kit).

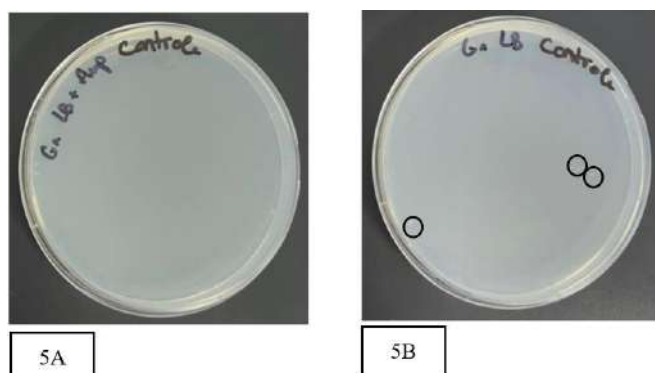


FIGURE 5 – RESULTS OF CONTROL OF BL21 WITHOUT PLASMID ON LB NUTRIENT AGAR WITH AND WITHOUT AMPICILLIN

(A) Results of culturing no transformed BL21 bacteria on petri with ampicillin. Bacteria no transformed from A. Leleu on LB agar + ampicillin.

(B) Results of culturing no transformed BL21 bacteria on petri without ampicillin. Bacteria no transformed from A. Leleu on LB. Circles show CFU (colony forming units).

Consequently, one application of the bacteria DH5- $\alpha$  transformed with nanobodies in it would be to express the nanobodies thanks to bacteria of expression (BL21). It is the end of the experiment that we showed in fig. 4 but by lack of time, we could not repeat the protocol. Moreover, due to this issue we could not express them directly to produce nanobodies and then purify them. However, this protocol should possibly work, and it could be interesting to do it again. Further studies could also be valuable to confirm that these nanobodies are efficient to detect GFAP by using them on brain slices.

Another interesting point of the method developed is that it does not require any animals. Indeed, most of the nanobodies produced today are made *in vivo* by using alpacas. To do so, the animal is immunized with the antigen of interest [14]. This first step only takes two months [14]. In comparison, our method *in vitro* here took one week and with the production of nanobodies, the whole process could take two weeks without any optimization. Outside of the gain of time that this allows, using animals to produce is very controversial today. Moreover, strict ethical regulations exist: the 3R strategy which has the principle to reduce animal use, refine their

existence and replace animal use, when possible, must be respected [15]. *In vitro* production techniques are also less expensive [14]. All these arguments are in favor of a democratization of our method.

## ACKNOWLEDGMENT

The success of this study would not have been possible without the unwavering support and contributions of our professor Dr. Grimaud, Jacqueline Bert, Ambre Leleu and CellTechs team. Their tireless efforts to educate and inspire their students have created a fertile environment for learning and discovery. Their passion for teaching and commitment to our success is deeply appreciated, and their contributions have played a significant role in the outcome of this study. We are grateful to have had the opportunity to work with such dedicated and talented educators and we would like to extend our sincerest thanks to the entire pedagogical team.

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# Development of a Method to Produce Nanobodies Against GFAP as a Promising Tool in the Field of Neurodegenerative Diseases

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**Abstract** – Understanding neurodegenerative diseases is a key issue in recent years. This includes research on the glial fibrillary protein, which is thought to be involved in these diseases. However, in order to study this protein, it is necessary to cross the blood-brain barrier, which is possible with nanobodies. But passing nanobodies through this barrier would not allow visualization of the protein of interest. Therefore, the nanobodies would have to be coupled to fluorescent proteins. So, is it possible to synthesize nanobodies combined with the green fluorescent protein directed against the glial fibrillary protein? This article aims to answer this question through various manipulations. To achieve the goal of creating a platform for the production of nanobodies against the glial fibrillary protein, several manipulations have been performed. It starts with the amplification of a plasmid, then the insertion of genes coding for nanobodies directed against the glial fibrillary protein coupled to the green fluorescent protein, and the expression of these genes through bacteria.

**Index Terms** – Glial Fibrillary Acidic Protein (GFAP), Green Fluorescent Protein (GFP), Nanobodies, Neurodegenerative diseases, Production platform.

## INTRODUCTION

Neurodegenerative diseases represent a major public health problem that is expected to increase with the aging of the population while the lack of treatment persists. A third of the French population is affected by a brain disorder such as Alzheimer's Disease (AD), with one person out of eight affected by this pathology after the age of 80 in Europe, or Alexander Disease (AxD), a rare pathology [1] [2].

A neurodegenerative disorder is characterized by memory loss and the decrease of intellectual functions [1] [2]. AD is the

most common cause of dementia in the world [3]. The main neuropathological characteristics are amyloid plaques and neurofibrillary tangles [3] [4]. Aberrant proteins can be observed in patients' brains, like Amyloid- $\beta$  (A $\beta$ ) and hyperphosphorylated tau [3].

Another rare neurodegenerative disease with multiple clinical forms ranging from new-born to adult is AxD divided into type I and II [2] [5] [6]. This pathology mainly affects the central nervous system (CNS) and is characterized by an abnormal or destructive development of the myelin sheath [7]. This disorder involves the development of megalencephaly, which is an increase in brain volume also accompanied by spasticity (increased muscle tone), and gradual mental retardation [2]. The disease is progressive, genetic, and metabolic. Most patients die within ten years of onset [8].

Hence, in numerous neurodegenerative diseases, the Glial Fibrillary Acidic Protein (GFAP) seems to play a key role [2] [3] [5] [6] [7] [8] [9]. Discovered in 1969 by Dr. Lawrence F. Eng, GFAP is a protein classified in the type III intermediate filament, which is encoded by the human gene 17q21 whose localization is on the chromosome 17 [10] [11] [12] [13]. It has a size of 432 amino acids and a molecular mass of 49,880 Daltons [11]. This protein is strongly expressed by cells of the CNS, mainly in the glial cells called astrocytes or in ependymal cells and neural stem cells [12] [13]. This is one of the main constituents of astrocytes and therefore of the CNS [13] [14]. Indeed, these proteins bind together to form intermediate filaments present in astrocytes [10] [11] [12] [13] [14]. In the peripheral nervous system, it is also present in other types of cells such as Schwann cells, enteric glial cells and satellite cells [12] [13] [15].

Currently, the role of GFAP is not entirely known [13] [14]. It is the main component of astrocytes that play a role in supporting and sustaining brain and spinal cord cells [7] [14].

In particular, astrocytes participate in the proper functioning of nerve cells, notably by sustaining oligodendrocytes and ensuring the presence of myelin, an essential substance for protection of nerves and efficient electrical conduction [14]. Thus, because of the localization of its expression, GFAP is involved in support, gliosis, and homeostasis [7] [14]. This protein is supposed to influence the form, motion and function of astrocytes and to provide strong support for the cells. It interferes in modulating mobility, cell proliferation, vesicle trafficking, and astrocyte-neuron interactions [7] [13]. Moreover, many types of brain tumours, probably derived from astrocytes [12]. If cells are damaged after a trauma or a disease, astrocytes respond by rapidly producing more GFAP [7] [14]. Thus, if its expression increases in the brain, there is a high probability of developing neurodegenerative disorders [7]. Furthermore, astrocytes help maintain the Blood-Brain Barrier (BBB) that limits the passage of substances between the blood and the brain [14].

With the identification of the role of GFAP in the studies, researchers have deepened its link in neurodegenerative diseases [7]. Pathological deposits of A $\beta$  protein and the presence of astrogliosis are important in the brain of patients suffering from AD and it reflects the level of gravity of this pathology [2] [3] [5]. Studies have shown a strong correlation between the density of reactivated astrocytes associated to neuritic plaques (NP) and the upregulated expression of GFAP [2] [3] [5]. Reactive astrocytes are found around and inside diffuse and compact plaques, but the increase of GFAP expression depends on the compact NP [16]. Besides, the level of GFAP mRNA highly increases with the augmentation of the density of A $\beta$  protein [16]. Furthermore, the level of GFAP in AD increases with the quantity of tangles burden and the disease duration [17]. Hence, GFAP is not dependent on AD, but it is involved in different cognitive functions and its presence in the brain is not necessarily pathological [7].

Heterozygous false sense mutations in the GFAP gene can cause AxD [2] [8]. Many patients have mutations that affect the amino acid arginine, which is replaced by cysteine [4] [9]. The overexpression and alteration of the GFAP causes protein aggregates in the cytoplasm of astrocytes, which then induces a change in neurons and other cells of nervous tissue [6] [7]. Accumulation of this mutated protein leads to the formation of astrocyte inclusion bodies, called Rosenthal fibres, along the CNS [18] [19]. How these fibres are involved in the pathogenesis of the disease is unknown. Patients with AxD are accompanied and cared for in order to improve their quality of life, as there is only symptomatic and supportive treatments [18] [19]. Nevertheless, the attention of therapists is focused on decreasing the expression of the GFAP gene [8]. Anti-Sense Oligonucleotides (ASO) seem to be an effective mean of decreasing or suppressing GFAP [7]. ASOs allow astrocytes to return to near-normal levels within a few weeks after injection [7]. As we have seen, GFAP is implied in major brain pathologies such as AD and AxD.

Currently, different tools exist to study GFAP or brain disorders [20]. Notably, anti-GFAP antibodies are very useful as biomarkers of astrocyte cells and then to distinguish several cell types, particularly for detecting certain tumours, but also other cognitive pathologies or trauma [15]. Indeed, this protein and its derivatives are highly expressed in the event of trauma or stroke [7] [14]. Furthermore, alternative splicing of the GFAP gene have shown the variety of possible isoforms, studied in order to have a better understanding of the development of neurodegenerative diseases [20]. Today, roles of GFAP are studied *post-mortem* or *in vitro* in cells or organoids, but this context is not representative because there is no more cell activity [20].

In order to explore the roles of GFAP *in vivo*, one of the main difficulties is to access brain cells. To reach that goal, we should use a tool that can pass through the BBB [20]. Indeed, as BBB protects the brain, it prevents large proteins from passing through the brain, such as classical antibodies.

However, to do this, there is a revolutionary technique, the nanobodies (Figure 1) [21] [22] [23]. This new tool for microscopy has been discovered to assist in the identification of GFAP without staining and which can be directed exclusively to this protein with genetic modification.

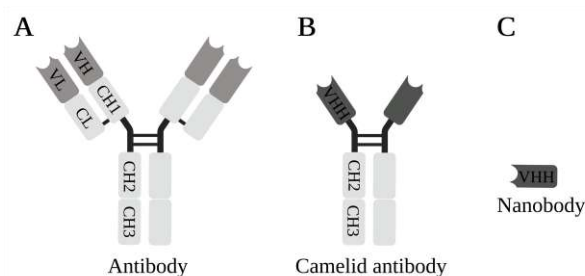


FIGURE 1 – STRUCTURE OF HUMAN AND CAMELID ANTIBODIES

(A) **Antibody.** Antibody is composed of two constant heavy chain regions (CH 1 - CH2 - CH3), two constant light chain regions (CL), two variable light chain regions (VL) and two variable heavy chain regions (VH).

(B) **Camelid antibody.** Camelid antibody is composed of two heavy chains (CH2 - CH3) and two variable domains (VHH).

(C) **Nanobody.** A nanobody is formed by a camelid antibody variable region fragment (VHH).

Nanobody or Variable Heavy-chain domain of camel Homodimeric antibodies (VHH) are different names describing the variable region of specific antibodies produced during an immune response by camels [24] [25] [26] [27]. Nanobodies were discovered in 1992 by Cecile Hamers-Casterman *et al.* [28]. These are specific antibodies observed in camelid serum that have a different structure than human antibodies. They are composed of a homodimer of heavy chains of the constant domain (CH2 and CH3) linked to a single variable domain, the antigen binding domain named VHH by a disulphide bond. Camelid antibodies lack the first

constant domain (CH1) unlike human antibodies [25] [26] [27] [29]. Indeed, human antibodies, like Immunoglobulin G (IgG), are composed of two heavy chains of the constant domain (CH1, CH2 and CH3) connected to two light chains (CL) via disulphide bonds and two variables region of the light chain and variables region of the heavy chain (VH and VL). Structurally, camelid antibodies also have a Y-frame like IgG (Figure 1) [30]. Nanobodies have many properties such as specificity, affinity, stability and solubility due to their hydrophobic side caused by the absence of VL fragment. They were also shown to remain functional at very high temperatures (90 °C) [31] [32]. Thanks to the small size of the VHH of only 15 kDa is the diffusion in the tissues and therefore a better efficiency which confers it interesting therapeutic properties [22] [32] [23]. For example, nanobodies can cross the BBB [33] [34]. Nanobodies are used in many fields of activity such as biotechnology, medical research, diagnosis or therapy in oncology, neurology, etc [26] [27] [33] [35] [36]. Specific nanobodies can be obtained via camelid immunization but more easily and cheaply via genetic engineering techniques [21] [37]. Indeed, the sequence of the nanobodies being known, it is thus possible to insert it in a plasmid and to carry out a bacterial transformation [38] [39]. It is also possible to modify the VHH sequence and target it against a particular antigen or to add a tag such as conjugating it to a fluorochrome [26]. In addition, an eGFP-bound nanobody can still cross the BBB [40]. That's why we would like to produce nanobodies against GFAP linked to enhanced Green Fluorescent Protein (e-GFP) and to use it as tools for microscopy. This project has many application areas, whether in basic research, in the clinic or for diagnostic purposes. In particular in basic research, our project could allow the quantification and localization of GFAP in tissues, for example by labelling astrocytes or any other cell type involved in neurodegenerative diseases.

## EXPERIMENTAL STRATEGY

### I. Works with E9-VHH anti-GFAP plasmid

The E9-VHH anti-GFAP plasmid contains an origin of replication, a resistance gene and a multiple cloning site cassette, which allows it to replicate independently of the bacterial chromosome, a selection pressure due to resistance to an antibiotic and the insertion of the gene of interest (Figure 2). The synthesis of the nanobodies is realized in four steps, the cloning of the gene of interest, the insertion of the recombinant plasmid in the competent bacteria, the culture of the recombinant bacteria and finally the synthesis of these proteins and its purification. After being cultured, the bacteria DH5- $\alpha$  were made competent with CaCl<sub>2</sub>. The CaCl<sub>2</sub> gave the bacteria the ability to incorporate DNA while the ice froze the membranes of the bacteria. The plasmid DNA stuck to the bacterial membranes and the passage of the bacteria at 42°C restored the fluidity of the bacterial membranes which led to the incorporation of pET-23d(+) E9-VHH anti-GFAP. Bacteria were then cultured to allow expression of the newly

incorporated resistance gene and then plated on LB medium with ampicillin to create the selection pressure to avoid plasmid ejection.

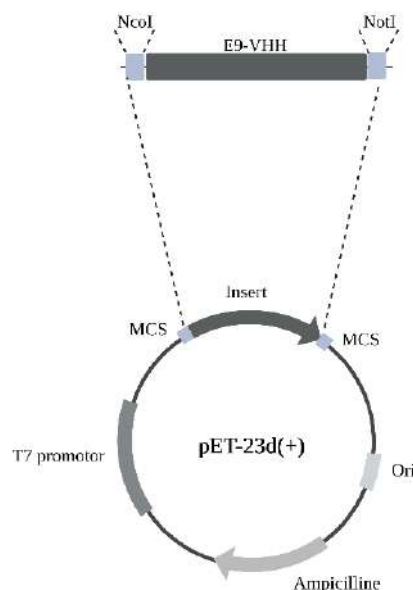


FIGURE 2 – E9-VHH ANTI-GFAP SEQUENCE\* AND PLASMID CONSTRUCTION RESULTS

Schema of the constructed insert containing a E9-VHH anti-GFAP sequence and two sites of restriction enzymes (NcoI, NotI). The digested size of the insert is 349 bp. The digested size of the plasmid is 3595 bp. The complete construct length is 3944 bp. GFAP: Glial Acidic Fibrillary Protein; VHH: Variable Heavy Chain fragment of a camelid antibody known as nanobody. \*Provided thanks to Pasteur Institute.

### II. Works with E9-VHH anti-GFAP linked to e-GFP plasmid

The E9-VHH anti-GFAP linked to e-GFP plasmid is composed by an origin of replication, a resistance gene and a multiple cloning site cassette, because we used the same backbone as the previous plasmid (Figure 3). We used the same strategy as for the first plasmid, but we did by ourselves the genetic engineering part. BL21(DE3) competent bacteria were transformed by the ligation product. The incorporation of pET-23d(+) E9-VHH anti-GFAP linked to e-GFP was done thanks to a heat shock. Bacteria were then cultured as it was described in the previous section.

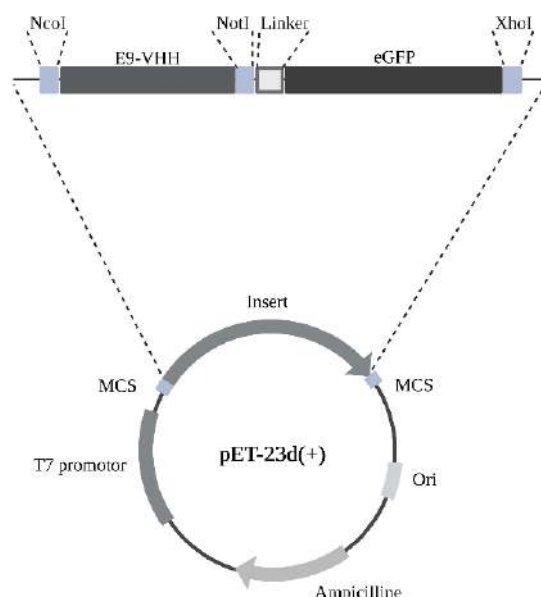
## MATERIALS & METHODS

### I. Works with E9-VHH anti-GFAP plasmid

#### a. pET-23d(+) containing E9-VHH anti-GFAP

The plasmid pET-23d(+) was provided by principal investigation (PI) Lafaye (Pasteur Institute) and is composed of E9-VHH insert [40]. In these sequences, there were several restriction enzyme sites such as NcoI or NotI that will facilitate the digestion step to separate the plasmid from the insert (Figure 2).





**FIGURE 3 – E9-VHH ANTI-GFAP LINKED TO E-GFP SEQUENCE AND PLASMID CONSTRUCTION**

Schema of the constructed insert containing a E9-VHH anti-GFAP sequence linked to a sequence of e-GFP. The linker added between these sequences ensures the correct folding of the resulting proteins of E9-VHH and e-GFP. Three sites of restriction enzymes (NcoI, NotI, and XhoI) and several random nucleotides are present in order to simplify the detection of digestion sites. The total size of the insert is 1115 bp. Nucleotides are added at each extremity to reach a length of approximately 1200 bp.

### b. Medium preparation

We prepared the Lysogeny Broth (LB) agar medium, it was composed of 10 g/L peptone (70175 Sigma-Aldrich), 5 g/L yeast extract (70161 Sigma-Aldrich), 10 g/L NaCl (provided by the Sup'Biotech teaching laboratories), 15 g/L agar (A5306 Sigma-Aldrich) and supplemented by 1 L of distilled water. Two types of mediums were prepared with or without ampicillin 100 µg/mL (provided by the Sup'Biotech teaching laboratories). To sterilise the mediums, we autoclave it during 30 minutes at 120 °C.

### c. Cells transformations

The bacterial transformation was performed on the *Escherichia coli* (*E.coli*) DH5- $\alpha$  strain provided by the Sup'Biotech teaching laboratories. This strain had allowed to select the desired gene. Once the bacterial growth had reached an Optical Density (OD) between 0.4 and 0.5. The transformation was done by CaCl<sub>2</sub> (0.1 M) to impermeabilize the cell walls. The bacteria were transformed by heat shock with 5 µL of plasmid. Transformed bacteria were spread on petri dishes containing LB agar medium and incubated overnight at 37 °C.

### d. Purification and selection of the clones

Minipreps were done (D4210 ZymoPur™ Plasmid Mini prep kit) to amplify the plasmid. A sterile LB tube with 100 µg/mL of ampicillin was inoculated by one colony from the petri dishes and was incubated overnight at 37 °C. A double

digestion was done with 1 µL of NcoI (C'CATGG) and 1 µL of NotI (GC'GGCCGC) restriction enzyme on the purified plasmid to verify the presence of the insert inside the plasmid on an agarose gel 1 %. We followed the manufacturer protocol to perform the double digestion (R0193S and R0189S NEB) [41]. A band was expected around 3500 bp for the plasmid and for the insert a band was expected at 400 bp (Figure 2).

### e. Material conservation

Bacteria rendered competent by CaCl<sub>2</sub> (0.1 M) and bacteria transformed by E9-VHH anti-GFAP were stored in tubes of 100 µL containing 15 % glycerol at a temperature of -80 °C.

Purified plasmid was stored directly in his tube at -20 °C.

## II. Works with E9-VHH anti-GFAP linked to e-GFP plasmid

### a. Design of E9-VHH anti-GFAP linked to e-GFP

The design of the nanobody was done in collaboration with PI Lafaye (Pasteur Institute) [40]. To obtain the E9-VHH anti-GFAP linked to e-GFP, the synthesis of the nanobody sequence was entrusted to a biotechnology company (GenScript) (Figure 3).

### b. Construction of E9-VHH anti-GFAP linked to e-GFP plasmid expression vectors

E9-VHH anti-GFAP linked to e-GFP was directly digested by XhoI (C'TCGAG) and NcoI (C'CATGG) restriction enzyme (NEB, R0146S and R0193S). pET-23d(+) plasmid was digested by following double digestion manufacturer's protocol during 1 hour at 37 °C [40]. Digestions were stopped by heating at 80 °C to inactivate the enzymes. Linearised plasmid and insert were confirmed by electrophoresis on a 1 % agarose gel and then purified by gel extraction method with the Wizard® SV Gel and PCR Clean-Up System (Promega). A band was expected at 1100 bp for the insert and for the plasmid a band was expected around 3500 bp.

E9-VHH insert was cloned into pET-23d(+) vector (Figure 3). The protein was expressed as a fusion protein with e-GFP in the C-terminal end, allowing it to be visualized by imaging. A 1:3 ratio was used to ligate the pET-23d(+) plasmid to E9-VHH anti-GFAP linked to e-GFP with T4 DNA ligase (NEB, M0202S) over night at 16 °C. Ligation was stopped by heating at 65 °C during 10 minutes and was stored at -20 °C.

### c. Expression cells transformations

Bacteria *E.coli* specifically BL21(DE3) competent cells provided by PI Lafaye were chemically transformed by the obtained ligation product. Transformed bacteria from transformations were inoculated onto LB agar plates with 100 µg/mL ampicillin and grown at overnight 37 °C. A piece of each colony on the plates was taken and diluted in Deoxyribonucleic Acid (DNA) free water. Bacteria was inactivated at 80 °C for 15 minutes. The presence of a plasmid was confirmed by electrophoresis on a 1 % agarose gel. A

band was expected around 4700 bp for plasmid containing the insert. If a band was not observable by this method, the colonies were amplified in 5 mL of LB medium with 100 µg/mL ampicillin and grown overnight at 37 °C.

## RESULTS

### I. Works with E9-VHH anti-GFAP plasmid

The goal of the project being the creation of nanobodies. This E9-VHH gene is directly directed against GFAP (Figure 2). Thus, the synthesized protein will be able to target the GFAP in astrocytes. Once the DH5- $\alpha$  bacteria have been made competent and the plasmid containing the gene of interest has been inserted, we prepare the media.

#### a. Preparation of culture media

The culture media necessary for the culture of DH5- $\alpha$  bacteria were made. Liquid and solid LB media were prepared. Petri dishes were poured from these media including four petri dishes with addition of ampicillin and four petri dishes without ampicillin (Figure 4A). The addition of ampicillin is essential because it induces a selective pressure.

#### b. Results of the cultures of the transformed bacteria and selection of clones

After one night of incubation, we had petri dishes previously inoculated with recombinant E. coli DH5- $\alpha$  bacteria. One of the two plates with LB medium was found to be colony-free. No colonies had grown on it (Figure 4A1). However, the petri dish containing LB medium with ampicillin was covered with white colonies of small diameter. We therefore suspected the presence of E. coli DH5- $\alpha$  bacteria possessing the recombinant plasmid with the gene of interest (Figure 4A2). To ensure the presence of the recombinant plasmid we checked the genetic material of the colonies on electrophoresis gel. We also selected about ten clones from the petri dish in figure 4A1 that we cryopreserved in glycerol.

#### c. Miniprep and electrophoresis gel

About 20 colonies were inoculated into sterile LB medium and the whole was cultured and incubated overnight. The next day the minipreps were started. A miniprep allows the extraction of DNA in the form of plasmids from transformed bacteria. Once plasmid DNA was obtained, an electrophoresis gel was run.

The initial plasmid pET-23d(+) with the gene of interest E9-VHH was digested with the restriction enzymes NotI and NcoI. The initial pET-23d(+) plasmid from the Pasteur Institute containing the E9-VHH gene was tested in duplicate in wells 3 and 6 and the miniprep extraction products in wells 2 and 5 (Figure 4B). The control was negative; no bands were observed. For the product of the recombinant bacteria, a band with a molecular weight of about 2.7 kb between 3 and 2 kb is observed, which is possibly the supercoiled form of the pET-23d(+) plasmid from the recombinant bacteria. For pET-23d(+) digested with NotI/NcoI a band of molecular weight

between 3 and 4 kb appears as well as a second band below 0.5 kb. The molecular weight of pET-23d(+) is 3663 bp and the E9-VHH insert is 349 bp which corresponds to the band sizes observed on the electrophoresis gel.

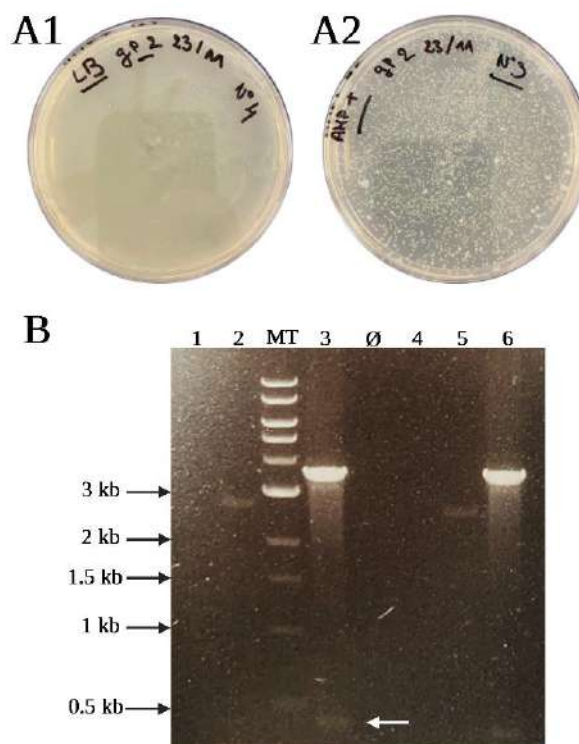


FIGURE 4 – AMPLIFICATION AND VERIFICATION OF THE E9-VHH VECTOR IN DH5- $\alpha$

(A) Results of the cultures of transformed bacteria on Petri dish with or without ampicillin. The first box was poured with only LB agar and seeded with transformed DH5- $\alpha$  bacteria (A1). The second box was poured with agar LB + Ampicillin and was then seeded with processed DH5- $\alpha$  bacteria (A2). The small white dots represent CFUs (Colony forming Units).

(B) Results of an electrophoresis gel. In wells 1 and 4 are negative control, 3 Kb band representing the supercoiled form of pET-23d(+) recovered by miniprep (2 & 5), the band between 3 Kb and 4 Kb represents the digestion of pET-23d(+) by NotI/NcoI enzymes and the band at 0.3 Kb represents the insert (3 & 6). MT well is a marker of size from 10 Kb to 1 Kb and the well with symbol Ø is an empty well.

### II. Works with E9-VHH anti-GFAP linked to e-GFP plasmid

#### a. Preparation of culture media

The culture media necessary for the culture of BL21(DE3) bacteria are liquid and solid LB media. We prepared two petri dishes with addition of ampicillin and two petri dishes without ampicillin. The addition of ampicillin is essential because it induces a selective pressure. Results of the cultures of the transformed bacteria and selection of clones

We inserted the recombinant plasmid in the competent bacteria BL21. In order to allow a good recovery of the

bacteria after the heat shock, we cultivated them for one hour under agitation in ultra-nutritive medium which was also given to us by the Pasteur Institute. After this incubation time, we saw our control tube clear and the tube with competent cells more turbid (Figure 5B). We took a few microliters from the tube containing the cloudiness with the competent bacteria BL21(DE3) to inoculate them on the petri dishes. One of the two plates with LB medium was found to be colony-free. No colonies had grown on it (Figure 5C1). However, the petri dish containing LB medium with ampicillin had only one white colony of small diameter. We therefore suspected the presence of BL21(DE3) bacteria possessing the recombinant plasmid with the gene of interest (Figure 5C2). To ensure the presence of the recombinant plasmid, we checked the genetic material of the colonies on electrophoresis gel.

#### b. Electrophoresis gel

We did a miniprep on the only colony found on the petri dish to extract the DNA present in the plasmid from our transformed bacteria. With that plasmid DNA obtained, we run an electrophoresis gel. The initial plasmid pET-23d(+) from DH5- $\alpha$  bacteria with the gene of interest E9-VHH was digested with the restriction enzymes NcoI and XhoI in order to add the new insert E9-VHH anti-GFAP linked to e-GFP (Figure 5A).

In well 1, we found the miniprep of DH5- $\alpha$  bacteria containing an undigested plasmid. That gave out the band between 5 and 6 kb and between 3 and 4 kb followed by a smear that corresponds to supercoiled, circular and linear forms of the plasmid. Well 2 corresponded to the miniprep product of DH5- $\alpha$  from the first part of the results digested by XhoI/NcoI which gave us the band between 3 and 4 kb. In theory the pET-23d(+) E9-VHH anti-GFAP plasmid is 3587 bp which was well observed in this well. In the wells 3 and 4, we found our designed insert that gave us bands between 2 and 3 kb which seems to be a contamination and a band between 1 and 1,5 kb that is in theory our insert because of its molecular weight of 1100 bp. The molecular weight of pET-23d(+) E9-VHH is 3587 bp which corresponds to the band size observed on the electrophoresis gel in well 2.

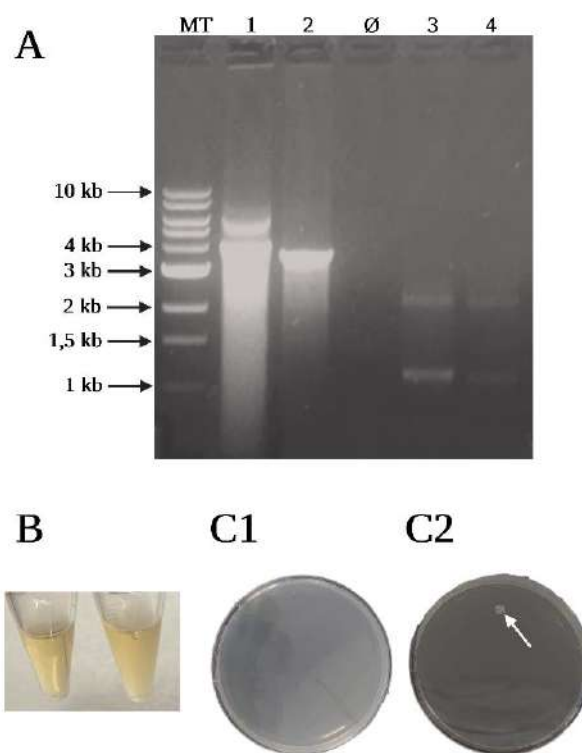
### DISCUSSION

Our manipulations were aimed at producing e-GFP coupled nanobodies directed against GFAP. To do so, we first multiplied the plasmid generously provided by the Pasteur Institute in DH5- $\alpha$ , then we inserted our E9-VHH anti-GFAP sequence in the plasmid, and finally we wanted to express this gene through BL21(DE3).

#### I. Works with E9-VHH anti-GFAP plasmid

First, we wanted to amplify the anti-GFAP E9-VHH plasmid from PI Lafaye by inserting it into DH5- $\alpha$ . The results of the agar plates were as expected, we had no colonies on agar without ampicillin and we had colonies on agar with

ampicillin (Figure 4A). We then checked that the plasmid contained the E9-VHH anti-GFAP insert, and then that DH5- $\alpha$  had integrated the plasmid (Figure 5A). We have a band around 3.7 Kb, which is a little bit above what was expected (3.5 Kb). A possible option is that it might be undigested plasmid. As during our double digestion, the result obtained was the expected one (a band at 3.5 Kb for the plasmid, and another under 0.5 Kb corresponding to our insert), we were certain to have the right plasmid. We therefore concluded that a simple digestion should have been done. The results of the electrophoresis confirm that the bacteria have integrated the plasmid and the E9-VHH anti-GFAP insert (Figure 5A). We can therefore conclude that multiplying the plasmid was a success.



**FIGURE 5 – VERIFICATION AND EXPRESSION OF THE E9-VHH ANTI GFAP LINKED TO E-GFP VECTOR IN BL21(DE3) BACTERIA**

(A) Results of an electrophoresis gel. MT well is a marker of size from 10 Kb to 1 Kb and the well with symbol Ø is an empty well. In wells are a pET-23d(+) undigested purified from BL21(DE3) bacteria: band between 6 and 5 Kb and between 4 and 3 Kb are supercoiled, circular and linear form of the plasmid (1), pET-23d(+) from DH5- $\alpha$  bacteria digested by XhoI/NcoI: band between 3 and 4 Kb is in theory the plasmid which is at 3587 bp (2), E9-VHH anti-GFAP linked to e-GFP insert digested by XhoI/NcoI: band between 2 and 3 Kb which is equivalent to a contamination and band between 1 and 1,5 Kb is in theory the insert which is 1100 bp (3 & 4).

(B) Results of competent cells. Competent cells then native pET-23d(+): control without insert, observation of a clear tube (1). Competent cells more pET-23d(+) then ligation product with E9-VHH anti-GFAP linked to e-GFP insert, observation of a turbid tube (2).

(C) Results of culturing transformed BL21(DE3) bacteria on Petri. Bacteria transformed by the anti-GFAP E9-VHH plasmid of the Institute Pasteur on LB agar + ampicillin (C1). Bacteria transformed by the plasmid E9-VHH anti-GFAP linked to e-GFP on LB agar + ampicillin (C2).

## II. Works with E9-VHH anti-GFAP linked to e-GFP plasmid

To build the plasmid, it was first necessary to remove the unwanted part of it by enzymatic digestion. The verification of this digestion was then checked on gel and confirmed us that this digestion was a success. On this same gel, we wanted to check that the insert we had ordered was the right one. We realized at that time that our insert was contaminated (Figure 5A). Indeed, during the construction of the insert, a DNase contamination occurred in the laboratory. This does not justify the presence of the unknown element since its size is 2.5 Kb whereas our insert is 1.1 Kb. If DNase had been involved, the contaminant would have been smaller than 1.1 Kb. Thus, we cannot say that it is DNase, but the fact that we had no contamination in the other wells suggests that it did occur during the making of the insert. The band of 2.5 Kb could be a plasmid in which the insert was produced by the company Genscript. As we verify the construction of PI Lafaye by doing the double digest by NcoI/XhoI, the plasmid should have been digested by NcoI/NotI to be sure that it was the pET-23d(+) E9-VHH anti-GFAP plasmid. Thus, we would have had two bands: one band for the plasmid and one band for the E9-VHH insert.

We then performed a ligation to couple the plasmid to our insert. The concern here is that due to the lack of time and volume, we could not perform a ligation control. So, we have no way to make sure that the ligation was done correctly.

The use of BL21(DE3) allows the expression of genes and were already competent. So, we just had to add the insert to get the plasmid recombinant. We restarted the culture by putting our bacteria in nutrient medium making a control with the native plasmid without insert. After one hour, a cloudiness was observed for the tube with plasmid and insert (Figure 5B). Based on the results, we thought that our bacteria had integrated the plasmid well. So, we continued the manipulations and placed our tubes on two agars (Figure 5C). It appeared that we had only one colony on the agar with our insert. This result is not normal because we expected to have a bacterial mat. However, during our manipulations, we did not have enough competent bacteria to put in the tube due to material constraints. During the gel analysis, it turned out that this colony did not have our plasmid. Therefore, we concluded that the re-culture step in nutrient medium with ampicillin did not allow the growth of the bacteria that had not integrated the plasmid since there was a selective pressure. Then, we could not express the protein as expected.

### PERSPECTIVES

The study reported here is continuing by incoming Sup'Biotech students. The aim of the incoming student is to achieve the originally intended purpose we have set in the beginning, which was to visualize the fluorescent antibodies linked to the GFAP with light sheet fluorescent microscopy. We have left materials kept at -80 °C that could be useful to realize the next steps [33] [34].

The main advantage of this technology is that it is more efficient than 2D models that has limitations. Cell Techs laboratory has developed mini brains. They have developed as well in a partnership with the French Alternative Energies and Atomic Energy Commission (CEA), a light sheet fluorescence microscopy that allows 3D photography of the mini brains created in their laboratory. If our experiment concerning the design of nanobodies linked to e-GFP directed against GFAP would be successful, it eventually could be tried with light sheet fluorescent microscopy to observe the presence of GFAP in brain organoids [42] [43]. The researchers of Pasteur Institute that have already worked on that topic before were very helpful to us. They supported us with their knowledge by answering all our questions about the different steps of the protocol and by providing us with some missing materials for the experiment for free. We recommend to the students of the following classes or the scientific community intending to work on this project, to maintain such a relationship with a reputed research institute such as Pasteur Institute and to believe that this collaboration will lead to successful results.

### CONCLUSION

Nanobodies, small antibodies that can cross the BBB, are promising elements for the therapy of neurodegenerative diseases. Indeed, the number of people suffering from neurodegenerative diseases is considerable and is constantly increasing. There would be a correlation between brain trauma, neurodegenerative diseases and the GFAP protein present in astrocytes. Astrocytes are involved in the proper functioning of nerve cells and GFAP influences the shape, mobility and function of these astrocytes. The more brain cells are damaged, the more GFAP is produced by the astrocytes. A correlation has been found between the GFAP present and the density of NP in AD. While AxD is caused by a mutation in the GFAP gene, it is therefore of utmost importance to focus on this protein. To better understand the role that GFAP has in these diseases, our goal was to construct nanobodies directly directed against GFAP with a e-GFP fluorophore that can be visualized under microscopy. We successfully obtained DH5- $\alpha$  bacteria with the pET-23d(+) E9-VHH anti-GFAP plasmid containing the E9-VHH anti-GFAP gene but it did not contain the e-GFP gene. We therefore digested the pET-23d(+) plasmid and inserted a E9-VHH anti-GFAP linked to e-GFP gene to obtain the fluorescence allowing us to detect the nanobodies in the sample. However, it turns out that the BL21(DE3) bacteria did not contain the plasmid of interest. One of the culture media in which they were inoculated did not have ampicillin, the selective pressure could not be done and thus the bacteria could have ejected the plasmid at that time. Finally, we cryogenised the DH5- $\alpha$  bacteria containing the pET-23d(+) E9-VHH anti-GFAP plasmid. The next classes of students will then take up the

project where we left off to continue the production of nanobodies. Organoid samples will be available for direct testing of their nanobodies.

### ACKNOWLEDGMENT

We warmly thank PI Lafaye from the Pasteur Institute for helping us to establish our protocol, and for providing us material to carry out our manipulations. We would also like to thank our teacher, Dr Grimaud, for all the help and support he gave us throughout this project. Our thanks go as well to the Sup'Biotech laboratories and to all the people attached to them for their welcome and their precious help. We also thank the Cell Techs laboratory for their theoretical help and the advice they gave us for the reagent samples they provided us. Our thanks go to BioRender software that we used to build our figure.

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# Preliminary Study of the Optimal Growth Conditions for *Serratia Marcescens* Dedicated to Optimizing the Production of Prodigiosin for Scaling Up to Industrial Level

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**Abstract** - More and more molecules from bacteria with biomedical properties are being discovered. This is the case of prodigiosin which is synthesized by *Serratia Marcescens*. It is a bacterium of the genus *Serratia* belonging to the Enterobacteriaceae family. This bacterium produces a biomolecule with extraordinary properties: Prodigiosin. Its production at large scale is compromised by the difficulties to make the bacteria producing it. To determine the simplest and less expensive means of production, we tested several media to observe the growth of *S. marcescens* and quantify it spectrophotometrically. We also performed growth kinetics of the bacteria associated with prodigiosin production as a function of time. We performed statistical analyses via ANOVA method to determine the best growth conditions. *Serratia Marcescens* growth correlated with prodigiosin production. Based on the analysis of our results, we have shown that LMTao medium at 20°C and LB media at 20°C are the most optimal mediums for the growth of *S. marcescens* among the panel of culture media tested. Moreover, we have shown that this same medium seems to be the most interesting to produce prodigiosin.

**Index Terms** – Optimization, Production, Prodigiosin, *Serratia Marcescens*.

## INTRODUCTION

*Serratia Marcescens* is a bacterium of the genus *Serratia* belonging to the family Enterobacteriaceae [1]. It is a gram-negative, mobile by peritrichous cilia bacillus that can grow at temperatures ranging from 5 to 40 °C and pH ranging from 5 to 9 [2]. Most strains of *Serratia Marcescens* are resistant to several antibiotics like quinolone antibiotics and some beta-lactams antibiotics which can cause opportunistic infections in several places including the urinary tract, respiratory tract,

wounds, and eyes [3] [4]. The ubiquitous bacterium *Serratia Marcescens* has a peculiarity that makes it recognizable in our environment. It produces a red pigment called prodigiosin and is located mainly in humid areas such as bathrooms [3].

Prodigiosin has many biomedical properties [5]. Indeed, it allows the triggering of apoptosis in tumor cells and therefore induces a decrease in proliferation, invasion, and migration of these. The prodiginine family, to which prodigiosin belongs, has shown apoptotic activity against a group of cancer cell lines. This includes lung, colon, kidney, and breast cancers, with low cytotoxicity in non-cancerous cells [6] [7] [8] [9]. Another interesting feature of prodiginines is that the induction of apoptosis is independent of the p53 protein (a protein that is one of the main intracellular factors activating an apoptosis mechanism). This p53 protein is frequently mutated in cancer cells and in defective cells in apoptotic pathways, showing the prodigiosine as a good alternative against cancer cells. In addition, prodiginines are not affected by the multidrug resistance pumps examined. For these reasons, prodiginines and its derivatives are interesting candidates as a chemotherapeutic agent [10] [11]. Added to this, the prodigiosine has an anticancer effect by modulation of autophagy in melanoma cells. Effectively, it modulates mTOR-dependent autophagy (mammalian target of rapamycin), which plays a key role in the regulation of autophagy in melanoma cells. In addition, prodigiosin can also trigger stress of the endoplasmic reticulum and thus induce the autophagic death of glioblastoma cells (or brain tumors). However, the exact molecular targets and channels through which prodigiosin regulates autophagy in colorectal cancer cells remains unclear. Prodigiosin is a novel autophagy inhibitor that can increase the effect of chemotherapeutic drugs and potentially sensitize recalcitrant patients [12]. While the market for cancer treatment grew by 12% in 2021, reflecting a growing number of diagnostics and interest from

pharmaceutical companies, this biomolecule could pave the way for new possibilities [13]. This promising molecule can also be used in the food industry as an additive [5]. However, the production of prodigiosin remains complex and requires fine regulation of parameters. The molecule is stable in conditions that are still not standardized. It has no typical growth medium, nor optimum temperature. This makes its interest in marketing insufficient. Moreover, to date, no large-scale production has been undertaken in view of the necessary means.

We were initially interested in studying its growing conditions thanks to the available literature on this subject. We tested three different culture media and three different growth temperatures as well. Then, thanks to our results, we tried to identify the extraction medium that made it possible to obtain the most prodigiosin.

Therefore, the purpose of this article will be to determine how to optimize the growth of *Serratia Marcescens* and its production of prodigiosin. Another goal of this article will therefore be to determine if prodigiosin's production is related to the growth conditions of *Serratia Marcescens*.

## MATERIALS AND METHODS

In order to understand the method, it is important to explain the context in which this study took place. The topic of prodigiosin was imposed during our 3 years of engineering studies. We had to set up a plan of action and experiments in accordance with the policy of our school. Thus, certain choices were made according to the material and financial means at our disposal.

We only had one week to carry out our experiments. In this time, we chose to test three environments and three temperatures. We would have liked to be able to test the growth of *Serratia Marcescens* in co-culture and to analyze the impact of the luminosity as well as of the O<sub>2</sub> saturation. It would have been interesting to work in bioreactors to be able to regulate as many parameters as possible, but for budgetary reasons, this was not feasible. We therefore chose to work with Erlenmeyer flasks in order to be able to test as many conditions as possible.

We also made a review of the literature on the subject of prodigiosin to be able to set up our protocols. This was particularly useful in the choice of the wavelength when quantifying the growth of *Serratia Marcescens* and the quantification of prodigiosin. Moreover, thanks to the experience of our supervisors, we were able to orient ourselves when we hesitated or when we did not have information in the literature.

### I. Preparation of culture media for *S. marcescens*

The different culture media used are described in table 1.

TABLE I – COMPOSITION OF THE CULTURE MEDIA

| MEDIUM                   | COMPONENTS                                                               |
|--------------------------|--------------------------------------------------------------------------|
| Préculture medium        | 250mL of LMTao medium                                                    |
| Nutrient broth BN (g/L)  | Tryptone 15 g/L, papain soybean peptone 5 g/L, sodium chloride 5 g/L     |
| LMTao (g/L)              | Yeast extract 4 g/L, tryptone 4 g/L, mannitol 6,7 g/L, peanut oil 20 g/L |
| Glyceol/Peptone GP (g/L) | Glycerol 20 g/L, peptone 20 g/L, MgSO <sub>4</sub> 15 g/L                |

### II. Resuspension of the inoculum

From *S. marcescens* agar strain, a 5mL inoculum was prepared at 0.5 MacFarland. This inoculum was seeded in the preculture medium and then placed in an incubator at 30°C, with horizontal agitation (110 rpm) for 24 hours [15].

### III. Growing conditions

Temperature: program the incubator to the desired temperature (20°C, 25°C and 30°C).

Time: the culture time for each condition is 21h.

For all conditions, agitate the Erlenmeyer flasks and unscrew the caps slightly. Inoculate at 5% (volume of bacteria/total volume (50mL)) [15].

Study controls:

- Positive control: sampling of the preculture and spreading on agar plate to check specificity.
- Negative control: each culture medium (3) not inoculated.

### IV. Reading the absorbance with a spectrophotometer

We chose to work at 600nm. This wavelength allowed to characterize the growth of the bacteria [15] [16] [17] [18].

## RESULTS

We wanted to know which culture medium was more optimal for *S. marcescens* between the 3 media: LMTao, nutrient broth and glycerol peptone.

According to Figure 1A, we have three different graphs that show absorbance as a function of growth time. These three graphs differ depending on the temperature used (20°C, 25°C or 30°C). We did a blank for all culture conditions according to the different culture media used and the temperature. We can clearly observe that there is no optical density for all blanks made.

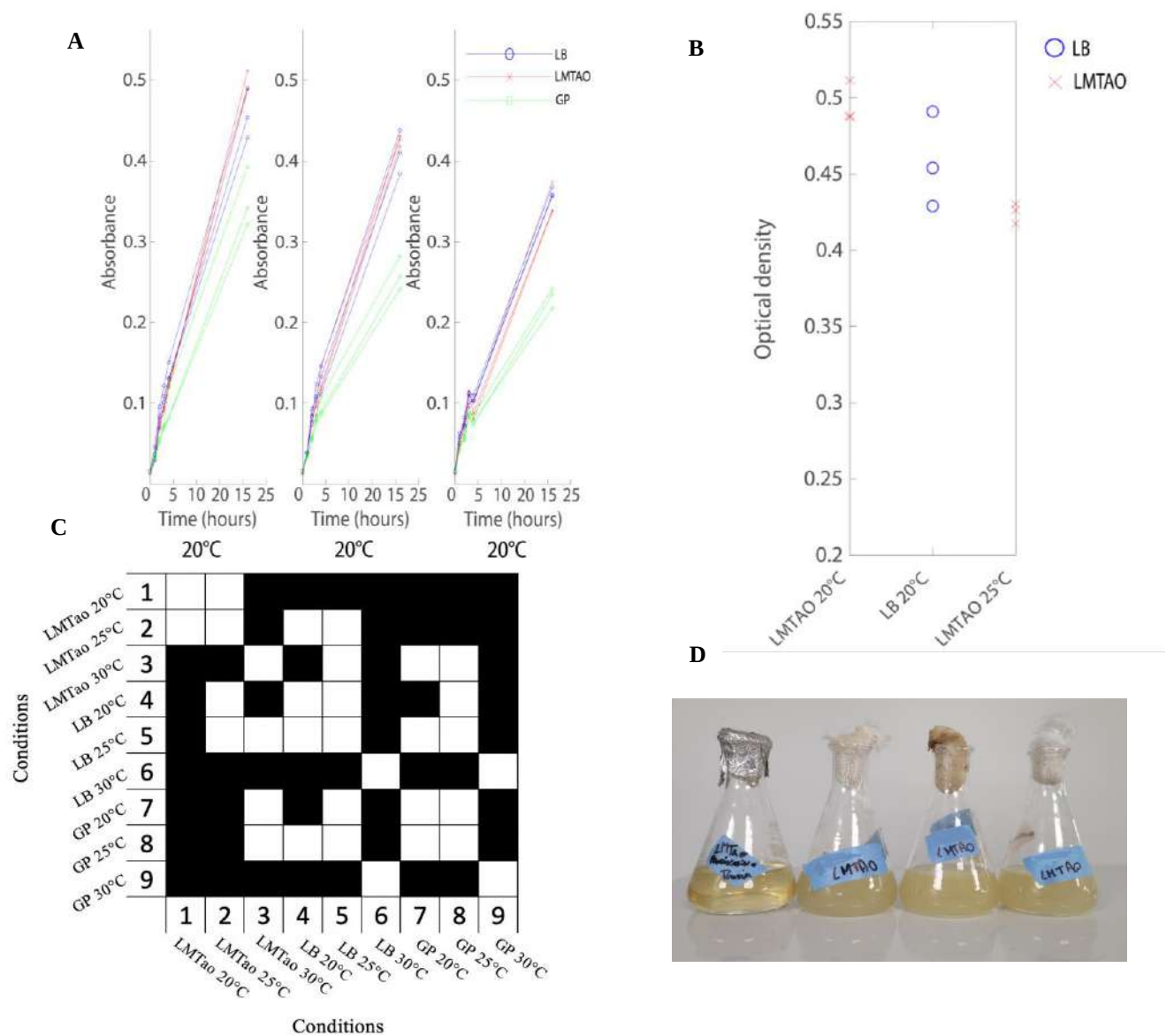


FIGURE 1– GROWTH COMPARISON OF *S. MARCESCENS* AT THREE DIFFERENT TEMPERATURES AND THREE MEDIA FOLLOWED BY STATISTICAL ANALYSIS.

- (A) Growth curve of *S. marcescens* at three different temperatures (20°C, 25°C and 30°C) for each media (LMTao, LB, GP).
- (B) Statistic comparison of *S. marcescens* growth for each parameter via ANOVA and Dunn-Sidak correction.
- (C) Visual Representation of ANOVA and Dunn-Sidak test for *S. marcescens* growth. This figure is the graphic representation of a post HOC test with Dunn-Sidak correction comparing the results of *S.marcescens* growth between the different media and temperature tested previously. The conditions are the following ones: (1) LMTao 20°C, (2) LMTao 25°C, (3) LMTao 30°C, (4) BN 20°C, (5) BN 25°C, (6) BN 30°C, (7) GP 20°C, (8) GP 25°C, (9) GP 30°C. White squares are associated to conditions where growth differences are not significative. Black squares are associated to conditions where growth differences are significative. Two conditions can only be compared if one of the two culture parameters are identical (rather temperature of media). Such conditions are called “compatibles”. If not, these conditions will not be compared. The compatible and significantly different conditions are the following ones: 1-3, 1-4, 1-5, 2-3, 2-8, 3-6, 3-9, 4-6, 7-9, 5-6, 8-9
- (D) Photography of inoculated erlenmeyers after 21h in LMTao media at 20°C. There is one erlenmeyer at the very left which corresponds to our negative control as it has not been inoculated. The three others are the LMTao erlenmeyers in triplicate.

For all media combined, we observe that the absorbances are the highest at 20°C. For the GP medium: at 20°C the three quantification curves are between 0.3 and 0.4 AU, at 25°C between 0.2 and 0.3 AU and at 30°C less than 0.3 AU. For the LB medium: at 20°C the three quantification curves are between 0.4 and 0.5 AU, at 25°C between 0.35 and 0.45 AU and at 30°C less than 0.4 AU. For the LMTao medium: at 20°C the three quantification curves are close to 0.5 AU, at 25°C between 0.40 and 0.45 AU and at 30°C lower than 0.4 AU. It is also observed that LMTao and LB media are the two media with the highest amount of *Serratia Marcescens* with a maximum absorbance at about 0.5 AU for these two conditions.

According to Figure 1B, we observe that the absorbance values are slightly better for LMTAO 20°C than LB 20°C, which agrees with the analysis of Figure 1A, with a maximum absorbance at 0.51 AU respectively, while the GP medium shows a maximum at only 0.47 AU. On the other hand, the results seem to be more repeatable for the LMTao medium than for the BN/LB medium or the GP medium.

According to figure 1C, the significant conditions called compatible (black) in this ANOVA are: 1-3, 1-4, 1-5, 2-3, 2-8, 3-6, 3-9, 4-6, 7-9, 5-6, 8-9. Among these pairs of conditions and from the graph in Figure 1, the conditions in each pair with the best growth of *S. marcescens* are the following conditions: 1, 2, 3, 4, 7, 6, and 8. By cross-checking with the pairs 1-3, 1-4, 1-7, 3-6, 2-8; conditions 1 and 2, corresponding to the LMTao medium 20°C and 25°C respectively, are the two conditions with the highest significant growth compared to the others (but these 2 conditions are not significantly different according to this ANOVA test).

According to figure 1D, in the picture of the Erlenmeyer flasks, we can observe the negative control without *Serratia Marcescens* which shows no trouble in the media and therefore no bacteria contamination. We can observe a cloudiness for the second Erlenmeyer flask that we inoculated with the bacterium. We have a growth of *Serratia Marcescens* in the LMTao medium at 20°C.

Therefore, as our negative control is still sterile, all the experiments were successfully performed without any contamination from an external source.

Our goal is to optimize *S. marcescens* growth to see if the prodigiosin production will be impacted. So, our objective was to quantify the amount of prodigiosin in each culture media for each temperature. The results are presented in the Figure 2.

According to Figure 2A, for all media combined, we observe that the absorbances are the highest at 20°C. For the GP medium: at 20°C the three quantification curves are between 0.4 and 0.5 AU, at 25°C between 0.3 and 0.4 AU and at 30°C

less than 0.3 AU. For the LB medium: at 20°C the three quantification curves are between 0.5 and 0.65 AU, at 25°C between 0.45 and 0.55 AU and at 30°C less than 0.5 AU. For the LMTao medium: at 20°C the three quantification curves are between 0.5 and 0.65 AU, at 25°C between 0.45 and 0.55 AU and at 30°C lower than 0.5 AU. It is also observed that LMTao and LB media are the two media with the highest amount of prodigiosin with a maximum absorbance at about 0.65 AU for these two conditions.

According to Figure 2C, the significant compatible (black) conditions in this ANOVA are: 1-3, 1-5, 2-3, 2-8, 3-6, 3-9, 4-6, 7-9, 5-6, 8-9. The results of these tests are the same as those obtained via the ANOVA in Figure 1 except for the conditions LMTao 20°C and LB 20°C (respectively conditions 1 and 4) where no significant difference is observed.

According to Figure 2B, we observe that the difference between LMTAO 20°C and LB 20°C is hardly visible, which agrees with the analysis of Figure 2C, with a maximum absorbance at 0.61 and 0.62 AU respectively, while the GP medium shows a maximum at only 0.47 AU. On the other hand, the results seem to be more repeatable for the LMTao medium than for the BN/LB medium or the GP medium.

According to Figure 2A and 2C, *S. marcescens* showed the highest production of prodigiosin in LB and LMTao media at a temperature of 20°C compared to all the conditions tested. In order to discriminate between these two conditions LMTao 20°C and LB 20°C, we carried out a statistical analysis test which allowed us to show a significant difference.

Among all the conditions tested in figure 2B, the ANOVA test does not allow to observe a significant difference between the conditions LMTAO 20°C and LB 20°C. As these two conditions are not significantly different from each other, neither of these two conditions can be excluded to produce prodigiosin.

These results show that the LMTao medium seems to be the most interesting to produce prodigiosin due to a maximum absorbance at 0.61 AU and a good repeatability of the results, however, this cannot exclude the LB medium which presents a high absorbance of prodigiosin at 0.62 AU, even if the results for this medium are less homogeneous.

The quantity of prodigiosin produced seems to match with the growth of *Serratia marcescens* in the culture media. Therefore, the next graph will allow us to determine if there is a correlation between these two parameters.

A correlation between the growth of *S. marcescens* as a function of medium and prodigiosin production is therefore established in Figure 3. This linear regression is used to see if there is a linear relationship between our two data sets: the growth of *S. marcescens* quantified at 620 nm and the amount of prodigiosin quantified at 500 nm at 21h of culture for all media and all temperatures.



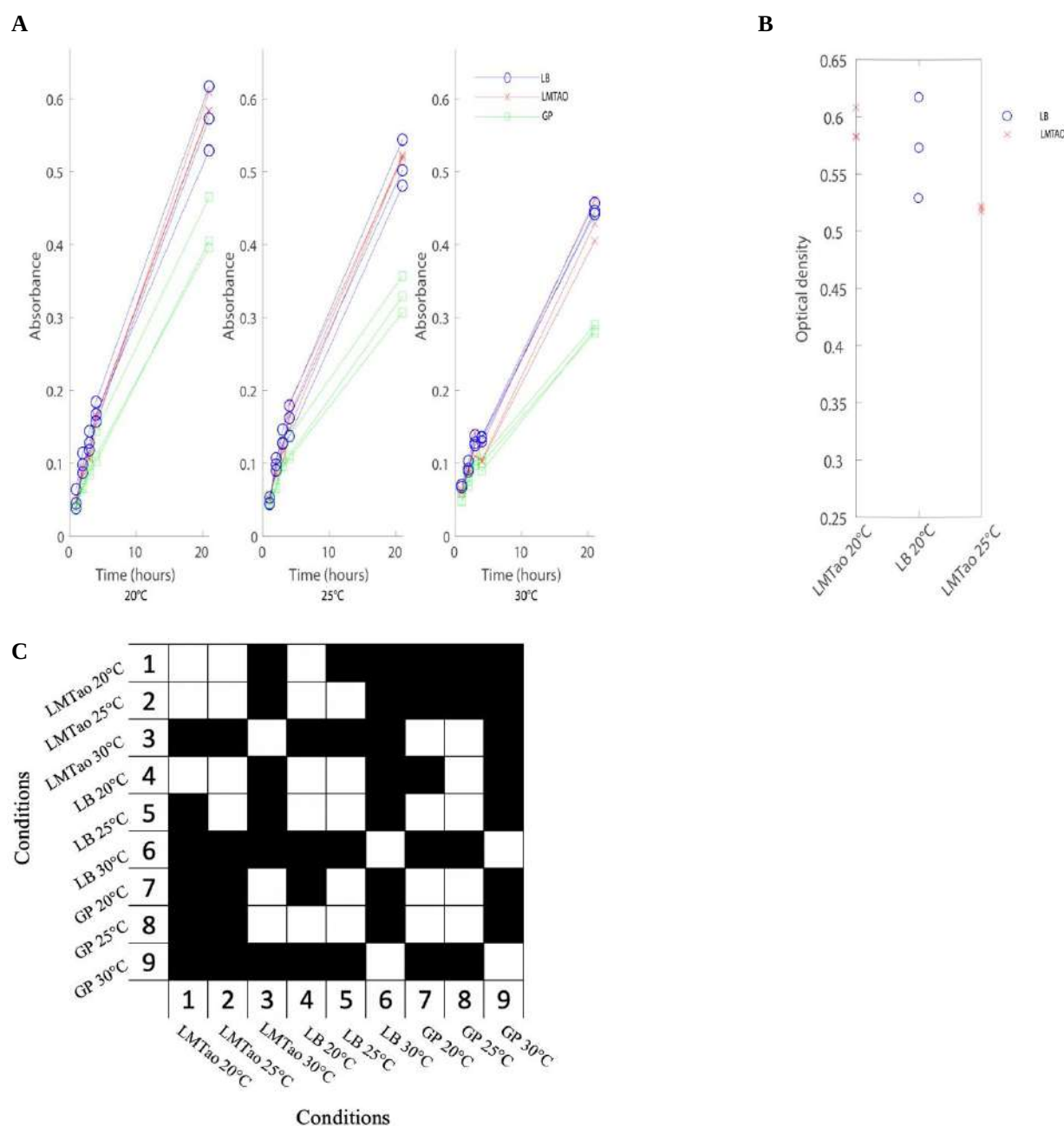


FIGURE 2 – PRODIGIOSIN PRODUCTION AT DIFFERENT TEMPERATURES, FOR EACH MEDIA (LMTao, LB, GP).

- (A) Statistic comparison of Prodigiosin production for each parameter via ANOVA and Dunn-Sidak correction.
- (B) Statistic comparison of *S. marcescens* growth for each parameter via ANOVA and Dunn-Sidak correction.
- (C) Visual Representation of ANOVA and Dunn-Sidak test for the Prodigiosin production. This figure is the graph representation of a post HOC test with Dunn-Sidack correction comparing the results of the production of prodigiosin between the different media and temperature tested previously. The conditions are the following ones: (1) LMTao 20°C, (2) LMTao 25°C, (3) LMTao 30°C, (4) BN 20°C, (5) BN 25°C, (6) BN 30°C, (7) GP 20°C, (8) GP 25°C, (9) GP30°C. White squares are associated to conditions where prodigiosin production differences are not significative. Black squares are associated to conditions where prodigiosin production differences are significative. Two conditions can only be compared if one of the two culture parameters are identical (rather temperature of media). Such conditions are called “compatibles”. If not, these conditions will not be compared. The compatible and significantly different conditions are the following ones: 1-3, 1-5, 2-3, 2-8, 3-6, 3-9, 4-6, 7-9, 5-6, 8-9.

If there is a linear correlation between these datasets, this indicates that prodigiosin production is correlated with *S. marcescens* growth.

To determine if there is a correlation, one must look at the coefficient of determination  $R^2$ . This is an indicator of the strength of fit between the regression model and the data collected. The closer this coefficient is to 1, the better the regression fits the data. Here our  $R^2$  is equal to 0.9995, which shows that the correlation between the variables is almost complete (and not completely, which is surely due to measurement uncertainties). We can therefore conclude that our first hypothesis is correct: there is a correlation between the growth of *S. marcescens* and the level of prodigiosin. We now know that to optimize the production of prodigiosin, we need to find the most suitable medium for the growth of *S. marcescens*.

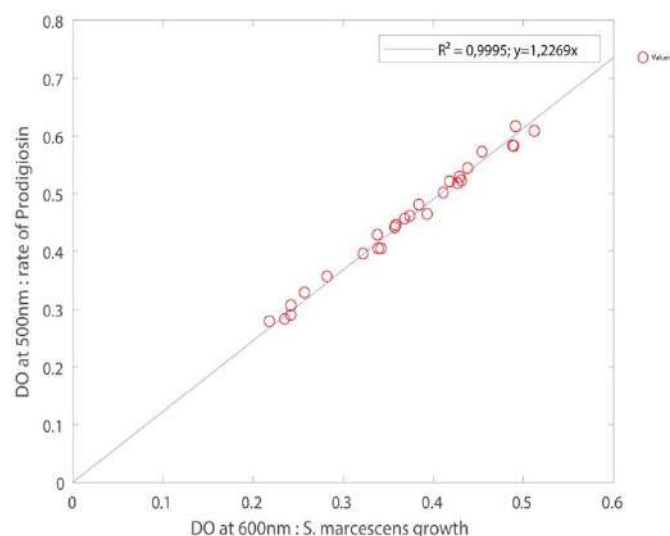


FIGURE 3 – CORRELATION BETWEEN THE GROWTH OF *SERRATIA MARCESCENS* AND THE PRODUCTION OF PRODIGIOSIN REPRESENTED IN A LINEAR REGRESSION

### CONCLUSION AND DISCUSSION

In this article, we have shown that the LMTao medium at 20°C is the most optimal medium for the growth of *S. marcescens* among the panel of culture media tested. In a second step, we have shown that it is still this same medium which seems to be the most interesting to produce prodigiosin. Even if the LB medium at 20°C seems to allow a production of prodigiosin relatively equivalent to this first medium. The linear regression curve for the two previous data sets shows us that there is a clear correlation between the growth of *S. marcescens* and the production of prodigiosin. Thus, we can affirm that under our experimental conditions, the greater the growth of *S. marcescens*, the greater the amount of prodigiosin produced.

In the continuity of these results, we were interested in the extraction of this prodigiosin produced in the LMTao

medium at 20°C. To do so, after lysis of the bacteria, we filtered the medium and tried a solvent/solvent extraction with a rotary evaporator [19] [20] [17]. However, the LMTao being composed of peanut oil and the prodigiosin not being soluble, the oil and the prodigiosin are found in the same phase and make the extraction complex and not feasible in our laboratory.

In view of all these results supported by a statistical study, we confirm that to obtain an optimum extraction of prodigiosin it would be preferable to find a medium equivalent or better than LMTao for the growth of *S. marcescens*, but which would have a much lower lipidic composition, such as LB medium. On the other hand, with more means and more time, it would be interesting to improve or develop with more attention this extraction protocol for the LMTao medium. Indeed, not having been able to extract the prodigiosin we can't rule on its functionality in the tested media, even if it is unlikely that the medium has an impact on our molecule of interest.

### ACKNOWLEDGMENT

We would like to thank M. Grimaud for the help he provided us to establish our protocol and its guidance and advice. We also would like to thank M. Favard for its help during our experiments, for having brought us his knowledge and advised us accordingly. We are also grateful to the supbiotech laboratories and those who work there for their help and kindness.

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# Sup'Biotech Projects

Volume 6  
February 2023

The Undergraduate Research Journal of Sup'Biotech

*Below are the cover designs that almost made it to the front page!*

